

Core Pure Mathematics 2

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Chapter 5: Polar Coordinates

Why this matters

Every curve you've drawn so far has used Cartesian coordinates (x, y) . Polar coordinates describe a point by its distance from a fixed origin (the pole) and the angle from a fixed direction (the initial line) instead — and this turns out to be the natural language for curves with rotational or spiral structure (cardioids, roses, spirals) that are awkward or impossible to describe neatly in Cartesian form. This chapter covers converting between the two systems, sketching polar curves, and finding areas enclosed by them using a new integral formula.

Key vocabulary

Term	Meaning
Pole	The fixed origin point (equivalent to $(0,0)$ in Cartesian)
Initial line	The fixed reference direction (equivalent to the positive x -axis)
Polar coordinates (r, θ)	r = distance from the pole, θ = angle measured anticlockwise from the initial line
Polar curve	A curve given by r as a function of θ , i.e. $r = f(\theta)$

Concept: converting between polar and Cartesian coordinates

A point with Cartesian coordinates (x, y) and polar coordinates (r, θ) are related by basic right-angled triangle trigonometry:

$$x = r \cos \theta, y = r \sin \theta, r^2 = x^2 + y^2, \tan \theta = y/x$$

To convert a Cartesian equation into a polar equation, substitute $x = r \cos \theta$ and $y = r \sin \theta$ throughout and simplify. To convert a polar equation into Cartesian form, use $r^2 = x^2 + y^2$ and $\tan \theta = y/x$, manipulating the polar equation until these substitutions apply.

Worked example 1 — Converting Cartesian to polar

WORKED EXAMPLE 1

Question. Find the polar equation of the circle $x^2 + y^2 - 4x = 0$.

Step 1. Substitute $x = r \cos \theta$ and $y = r \sin \theta$: **(M1)** $(r \cos \theta)^2 + (r \sin \theta)^2 - 4(r \cos \theta) = 0$

Step 2. Simplify using $\cos^2 \theta + \sin^2 \theta = 1$: **(M1)** $r^2(\cos^2 \theta + \sin^2 \theta) - 4r \cos \theta = 0 \rightarrow r^2 - 4r \cos \theta = 0$

Step 3. Factor out r ($r = 0$ is just the pole, contained in the curve for all θ , so divide through): **(A1)**

$$r(r - 4\cos \theta) = 0 \rightarrow r = 4\cos \theta$$

Answer: The polar equation is $r = 4 \cos \theta$.

EXAM TIP

Circles through the pole always simplify beautifully to $r = a \cos \theta$ or $r = a \sin \theta$ forms — recognising this pattern ("circle through the origin $\rightarrow r = a \cos \theta$ or $a \sin \theta$ ") lets you sanity-check your algebra: if you don't land on a clean $r = f(\theta)$, you've likely made an error.

Worked example 2 — Converting polar to Cartesian

WORKED EXAMPLE 2

Question. Find the Cartesian equation of the curve with polar equation $r = 2/(1 - \cos \theta)$.

Step 1. Multiply both sides by $(1 - \cos \theta)$: **(M1)** $r(1 - \cos \theta) = 2 \rightarrow r - r \cos \theta = 2$

Step 2. Substitute $r \cos \theta = x$, and isolate r : **(M1)** $r - x = 2 \rightarrow r = x + 2$

Step 3. Square both sides and use $r^2 = x^2 + y^2$: **(M1)** $r^2 = (x + 2)^2 \rightarrow x^2 + y^2 = (x + 2)^2 = x^2 + 4x + 4$

Step 4. Simplify: **(A1)** $y^2 = 4x + 4$, i.e. $y^2 = 4(x + 1)$

Answer: The Cartesian equation is $y^2 = 4(x + 1)$, a parabola.

Concept: sketching polar curves

To sketch $r = f(\theta)$, build a table of key values of θ (typically multiples of $\pi/6$ or $\pi/4$) and the corresponding r , and plot the points in order, watching for:

- values of θ where $r = 0$ (the curve passes through the pole);
- values of θ where r is negative — by convention, when $r < 0$, the point is plotted in the *opposite* direction (angle $\theta + \pi$), which can create loops;
- symmetry — if $f(-\theta) = f(\theta)$, the curve is symmetric about the initial line; if $f(\pi - \theta) = f(\theta)$, it's symmetric about the line $\theta = \pi/2$.
- For example, if a curve gives $r = -2$ at $\theta = 0$, you don't plot a point 2 units out along $\theta = 0$ — you plot it 2 units out along $\theta = 0 + \pi$ instead, i.e. at the Cartesian point $(-2, 0)$. Getting this backwards is a common cause of a sketch coming out as a mirror image of the correct one.
- **Tangents at the pole.** Every value of θ for which $r = 0$ gives you a tangent line to the curve at the pole itself, pointing in direction θ — as θ sweeps through that value, the curve passes through the origin heading in that exact direction. For the cardioid $r = 2(1 + \cos \theta)$ in Worked example 3, $r = 0$ only at $\theta = \pi$, so the curve touches the pole once, with tangent line $\theta = \pi$ there (matching the single cusp you can see in the sketch).

Key vocabulary — common named polar curves

Curve type	Typical equation	Shape
Circle through pole	$r = a \cos \theta$ or $r = a \sin \theta$	circle of diameter a , tangent to

		pole
Cardioid	$r = a(1 + \cos \theta)$ or $r = a(1 - \cos \theta)$	heart-shaped curve, one cusp at the pole
Limaçon	$r = a + b \cos \theta$ ($a \neq b$)	dimpled or looped curve depending on ratio a/b
Rose curve	$r = a \cos(n\theta)$ or $r = a \sin(n\theta)$	n petals if n odd, $2n$ petals if n even

Worked example 3 — Sketching a cardioid

WORKED EXAMPLE 3

Question. Sketch the curve $r = 2(1 + \cos \theta)$ for $0 \leq \theta \leq 2\pi$, indicating the value of r at $\theta = 0, \pi/2, \pi, 3\pi/2$.

Step 1. Build a table of key values: **(M1)**

θ	0	$\pi/2$	π	$3\pi/2$
$\cos \theta$	1	0	-1	0
$r = 2(1 + \cos \theta)$	4	2	0	2

(A1) — correct values

Step 2. Note the symmetry: $\cos(-\theta) = \cos \theta$, so the curve is symmetric about the initial line (the polar axis). **(B1)**

Step 3. Note $r = 0$ at $\theta = \pi$, so the curve passes through the pole exactly once, forming a single cusp there, and r is largest ($r = 4$) at $\theta = 0$. This is the classic heart/cardioid shape. **(M1)**

Answer: The curve is a cardioid symmetric about the initial line, with a cusp at the pole ($\theta = \pi$) and its widest point ($r = 4$) on the initial line at $\theta = 0$. (A sketch would show a heart shape opening towards $\theta = 0$, pinched to a point at the pole in the direction $\theta = \pi$.) **(A1)**

EXAM TIP

Physically sketching by hand, plotting 8–12 points around the full range of θ , does far more for your understanding than memorising named shapes. Always check for negative r (does the curve fold back on itself?) and for repeated values (does the curve retrace itself before θ completes a full 2π ?) — both are common exam sketch-marking points.

Concept: area enclosed by a polar curve

The area swept out by the radius vector as θ increases from α to β is found by summing infinitesimally thin "triangular" sectors of area $\frac{1}{2}r^2 d\theta$:

This is really just the GCSE circular-sector area formula in disguise. A full circle of radius r has area πr^2 , swept out over an angle of 2π , so a thin sector of angle $d\theta$ has area $(d\theta/2\pi) \times \pi r^2 = \frac{1}{2}r^2 d\theta$. Summing (integrating) these thin sectors from α to β gives the formula below — the only new idea is that r itself now changes as θ varies, instead of staying fixed.

$$\text{Area} = \frac{1}{2} \int(\alpha \text{ to } \beta) r^2 d\theta$$

This formula only gives the area of the region swept out as θ goes from α to β — for a full closed loop, integrate over the θ -range that traces that loop exactly once, and for the area *between* two curves, subtract one integral from the other over the overlapping range.

Key formula

KEY FORMULA

Area enclosed by $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$: $A = \frac{1}{2} \int(\alpha \text{ to } \beta) r^2 d\theta$

Worked example 4 — Area enclosed by a polar curve

WORKED EXAMPLE 4

Question. Find the area enclosed by the curve $r = 3\sin \theta$ for $0 \leq \theta \leq \pi$.

Step 1. Set up the area integral: **(M1)** $A = \frac{1}{2} \int_0^\pi (3\sin \theta)^2 d\theta = \frac{1}{2} \int_0^\pi 9\sin^2 \theta d\theta$

Step 2. Use the double angle identity $\sin^2 \theta = (1 - \cos 2\theta)/2$ to integrate: **(M1) — standard technique carried over from Core Pure/Core Maths trig integration** $A = (9/2) \int_0^\pi (1 - \cos 2\theta)/2 d\theta = (9/4) \int_0^\pi (1 - \cos 2\theta) d\theta$

Step 3. Integrate: **(M1)** $(9/4) [\theta - \frac{1}{2}\sin 2\theta]$ from 0 to $\pi = (9/4)[(\pi - \frac{1}{2}\sin 2\pi) - (0 - \frac{1}{2}\sin 0)] = (9/4)(\pi - 0) = 9\pi/4$

Answer: $A = 9\pi/4$. **(A1)**

EXAM TIP

This curve is actually a full circle of radius 1.5 traced out as θ runs from 0 to π (not 0 to 2π , since $r = 3\sin \theta$ retraces itself after π) — a lovely check is to compute the area as a circle directly ($\pi \times 1.5^2 = 2.25\pi = 9\pi/4$) and confirm it matches. Building this kind of cross-check habit pays off hugely for catching wrong integration limits.

Worked example 5 — Area between two polar curves (finding limits of intersection)

WORKED EXAMPLE 5

Question. Find the area of the region that lies inside both $r = 2$ and $r = 2 + 2\cos \theta$, for $0 \leq \theta \leq \pi$.

Step 1. Find where the curves intersect: set $2 = 2 + 2\cos \theta \rightarrow \cos \theta = 0 \rightarrow \theta = \pi/2$. **(M1)**

Step 2. For $0 \leq \theta \leq \pi/2$, compare which curve is closer to the pole (the smaller r bounds the "inside both" region): at $\theta = 0$, $r = 2$ (circle) vs $r = 4$ (cardioid) — the circle is smaller, so the circle $r = 2$ bounds the region here. For $\pi/2 \leq \theta \leq \pi$, at $\theta = \pi$, $r = 2$ (circle) vs $r = 0$ (cardioid) — the cardioid is smaller here. **(M1) — identifying the correct bounding curve on each sub-interval**

Step 3. Set up two separate integrals and add them: **(M1)** $A = \frac{1}{2} \int_0^{\pi/2} (2)^2 d\theta + \frac{1}{2} \int(\pi/2 \text{ to } \pi) (2 + 2\cos \theta)^2 d\theta$

Step 4. Evaluate the first integral: **(A1)** $\frac{1}{2} \int_0^{\pi/2} 4 d\theta = 2[\theta] \text{ from } 0 \text{ to } \pi/2 = 2(\pi/2) = \pi$

Step 5. Expand and evaluate the second integral, using $\cos^2 \theta = (1 + \cos 2\theta)/2$ (the same double angle identity as Worked example 4): $(2+2\cos \theta)^2 = 4 + 8\cos \theta + 4\cos^2 \theta = 4 + 8\cos \theta + 2(1+\cos 2\theta) = 6 + 8\cos \theta + 2\cos 2\theta$ **(M1)** $\frac{1}{2} \int(\pi/2 \text{ to } \pi) (6 + 8\cos \theta + 2\cos 2\theta) d\theta = \frac{1}{2}[6\theta + 8\sin \theta + \sin 2\theta] \text{ from } \pi/2 \text{ to } \pi = \frac{1}{2}[(6\pi + 0 + 0) - (3\pi + 8 + 0)] = \frac{1}{2}(3\pi - 8) = 1.5\pi - 4$ **(A1)**

Answer: Total area = $\pi + 1.5\pi - 4 = 2.5\pi - 4$.

Chapter summary

Idea	Key point
Conversion	$x = r \cos \theta$, $y = r \sin \theta$, $r^2 = x^2 + y^2$, $\tan \theta = y/x$
Sketching	Table of values; check symmetry, $r = 0$ points, and negative r
Named curves	Circles ($r = a \cos \theta$, $a \sin \theta$), cardioids, limaçons, rose curves
Area formula	$A = \frac{1}{2} \int r^2 d\theta$ over the θ -range tracing the region exactly once
Area between curves	Find intersection angles, identify the bounding curve on each sub-interval, integrate separately and add/subtract

Common exam mistakes

- Using r instead of r^2 in the area formula (forgetting to square before integrating).
- Forgetting the factor of $\frac{1}{2}$ in the area formula.
- Integrating over the wrong range of θ (e.g. 0 to 2π when the curve is only traced once over 0 to π), giving double the correct area.
- Not checking which curve is closer to the pole on each sub-interval when finding an overlapping area, leading to the wrong bounding function being used.
- Sign errors when converting $r \cos \theta \rightarrow x$ or $r \sin \theta \rightarrow y$, especially when rearranging polar equations with negative terms.

You should now be able to

- Convert between polar and Cartesian coordinates and equations in both directions.
- Sketch polar curves using a table of values, symmetry, and awareness of negative r .
- Recognise standard named polar curves (circles, cardioids, limaçons, rose curves) from their equations.
- Find the area enclosed by a polar curve, and the area of overlap between two polar curves, using $A = \frac{1}{2} \int r^2 d\theta$.